

The **tech** that powered



As we recover from the horror of the Mumbai attacks, we are now horrified by another thought: how did this gang get the technological one-up on our forces? And how must the good guys fight them?

however, this is considerably more complex.

Firstly, it isn't the satellite that is calculating the receiver's position, it's the receiver itself. Secondly, GPS doesn't need a satellite, it needs at least four. There are around 32 satellites in orbit dedicated to GPS, so finding four usually isn't a problem. Your receiver monitors the four strongest signals that it receives — signals that contain information about the satellite's own orbit, and the rough orbits of other satellites. It then uses the time it took for each message and calculates its distance from the satellite. Finally, using a method called *geometric trilateration* — for which it uses the orbital information for the satellites — it calculates its position on Earth.

Google Earth uses photos of the Earth taken by satellites, and slaps them on a virtual sphere. The result, as you've already seen, is a virtual world that you can tour on your desktop. Want to see a top view of Everest? It's there.

Agent 001

It's been all over the news — “Mumbai attacks reveal sophisticated use of technology”, “Gunmen used technology as tactical tool”, “Mumbai terrorists more tech-savvy than the police” — but while the mainstream media winds up portraying these terrorists as the evil geeks you only see in movies, or saying that the technology is to blame, the reality is that these terrorists just got creative. What they really did was to combine some simple tech, and use it to deadly effect.

The Tech:

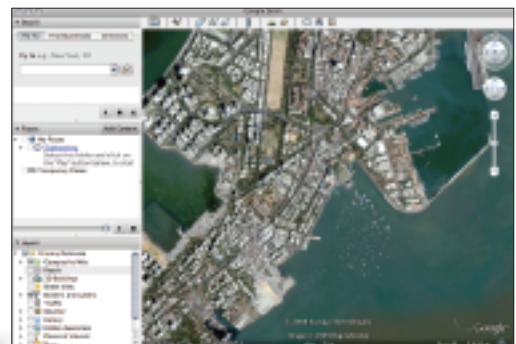
GPS and Google Earth

What they used it for:

Navigating their way to India, planning the attack, getting around Mumbai (likely).

How it works:

In concept, the Global Positioning System (GPS) is pretty simple. You ask your GPS receiver where you are, the receiver asks a satellite, the satellite sends back an answer, and there you go. In reality,



Fighting it

It's safe to expect that the Google-Earth-national-security issue will be dredged up again, and we wouldn't be surprised at a few demands to ban the software itself.

There might even be a similar outcry against GPS, but that won't last long. The bottom line is that both these technologies have caused more good than harm, and it would be a crying shame to limit them in any manner. GPS is an important tool on aircraft, ships, cars, and now even cell phones. And because a GPS device doesn't send any personal information to the satellites, it's impossible to tell who's using which device.

The smarter plan? Master both and use them to plan counter-terrorist activities.

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